

HSBC: Recharge Failure Recovery System

Developed an architecture using the circuit breaker pattern to ensure system resilience during external service outages. Key technologies used:

- AWS
- Kubernetes
- Microservices
- PostgreSQL
- Cassandra
- Kafka (asynchronous communication)

Problem Statement: external services were down and users were unable to do successful recharges

Solution: To solve this issue, we designed the system to show users the bill details and process the recharge request once the services became available.

Key Responsibilities

- Gathered and analyzed client requirements to define project scope and solutions.
- Collaborated in resource planning and allocation based on project needs.
- Designed and developed Spring Boot microservices for recharge failure recovery workflows.
- Implemented circuit breaker-based resiliency mechanisms to handle external service downtime.
- Participated in the complete SDLC, including development, testing, production deployment, and post-production support.

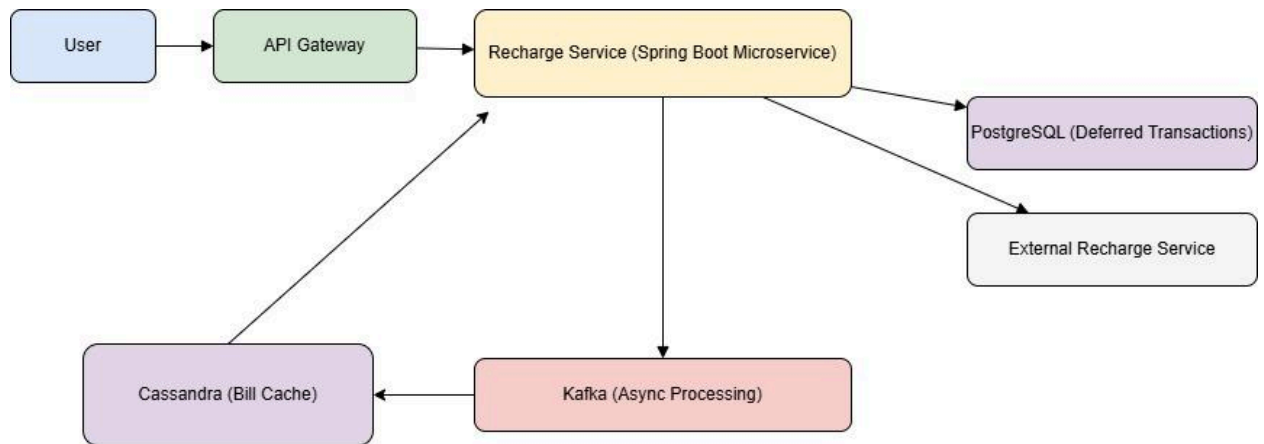
Three-Stage Recovery Process:

1. **Data Caching:** Bill details from external services are cached in Cassandra during normal operations.
2. **Failure Handling:** during failures, we fetched these bill details from Cassandra and stored the recharge requests in PostgreSQL in a deferred state.
3. **Recovery & Processing:** we validated whether the external service was up or not by sending a few test requests. Once the service became available, we started processing the deferred requests successfully.

Circuit Breaker States

State	Description
Closed	Normal operations with direct external service calls.
Open	External service failures occurred, so we avoided repeated external calls and used Cassandra data while storing recharge requests in a deferred state
Half-Open	we validated service recovery by sending a few requests to the external services

Architecture



Circuit-Breaker Diagram

